



ARCHITECTURE · EFFICIENCY · TRAINING STABILITY · PROVENANCE

从 $3 \times \text{GDN} + 1 \times \text{QSA}$ 的混合 token mixer，到四分支 Gated Residual、离卡 n-gram memory、Muon/Canzona 与稳定性压力测试：沿技术报告原有论证顺序，拆解这套 Qwen4 前瞻架构。

125B / 6B active

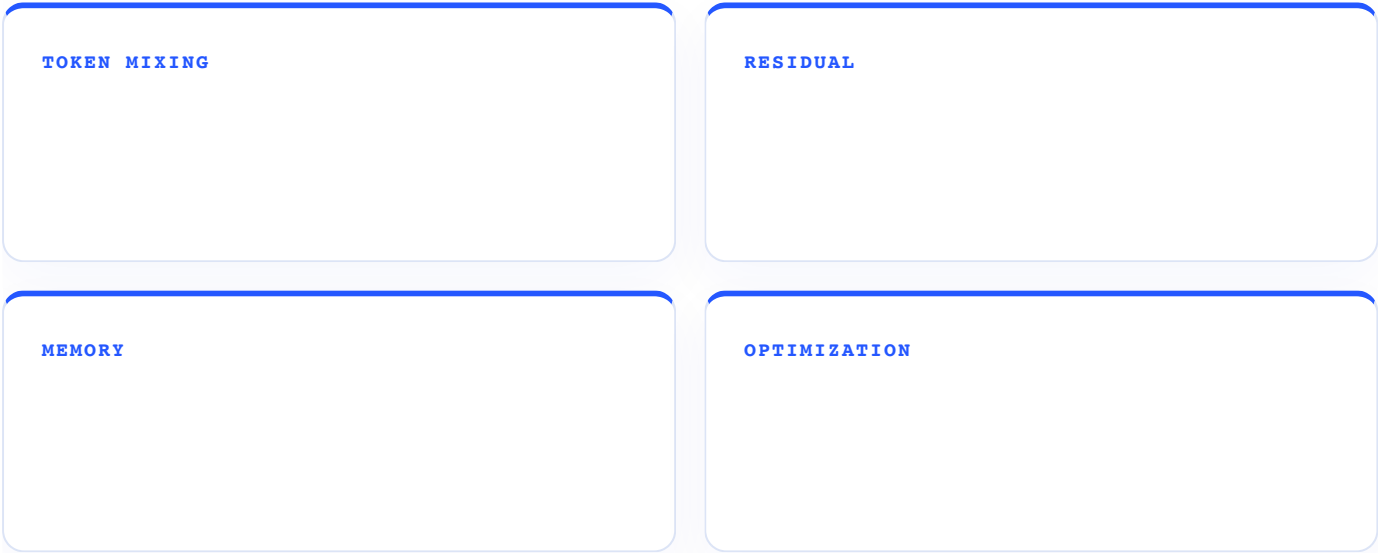
+51B n-gram

48 layers · 512 experts

256K native · 1M YaRN

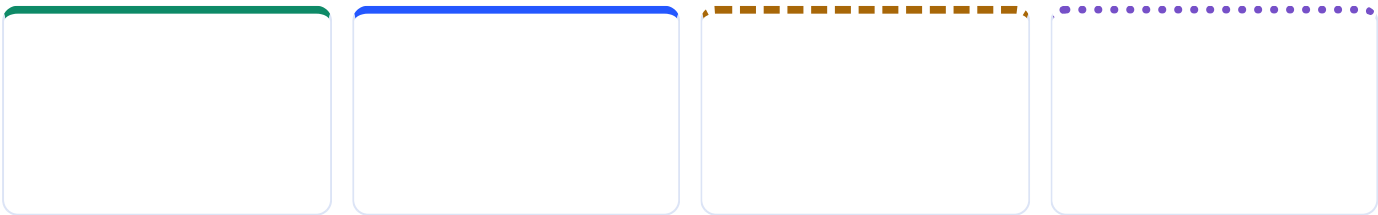
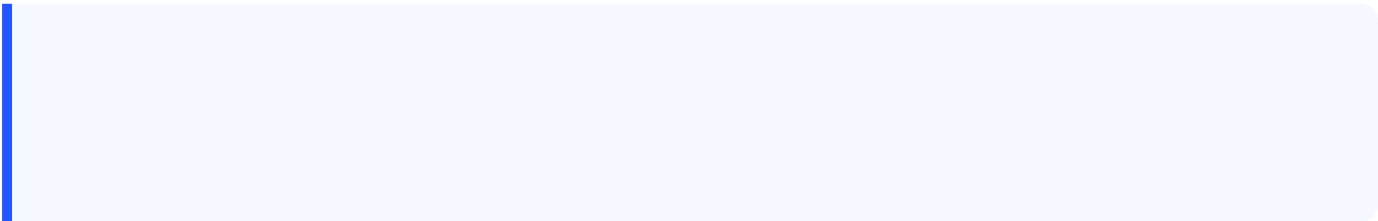
00 · EXECUTIVE READING

Qwen3.8-Flash-Next 的主线可以概括为：用 GDN 压低大多数层的序列状态成本，用 QSA 保留周期性的精确全局检索；用四分支 GR 改写跨层信息流并提供显式幅度调节；再以离卡 n-gram 表扩充“记忆容量”，最后重新拟合 Muon 下的批量、学习率与稳定性边界。



“Architecture, efficiency, and optimization form one coupled system.”

Qwen3.8-Next Technical Report, §5



01 · SCOPE & RELEASE AUDIT

技术报告主要研究文本主干的架构选择、效率和预训练稳定性；公开的 Qwen3.8-Flash-Next 权重则是一个已后训练的多模态模型。理解这一点，才能正确归属论文表格、官方模型卡 benchmark 和部署信息。

TECHNICAL REPORT

OFFICIAL RELEASE

02 · MACRO ARCHITECTURE

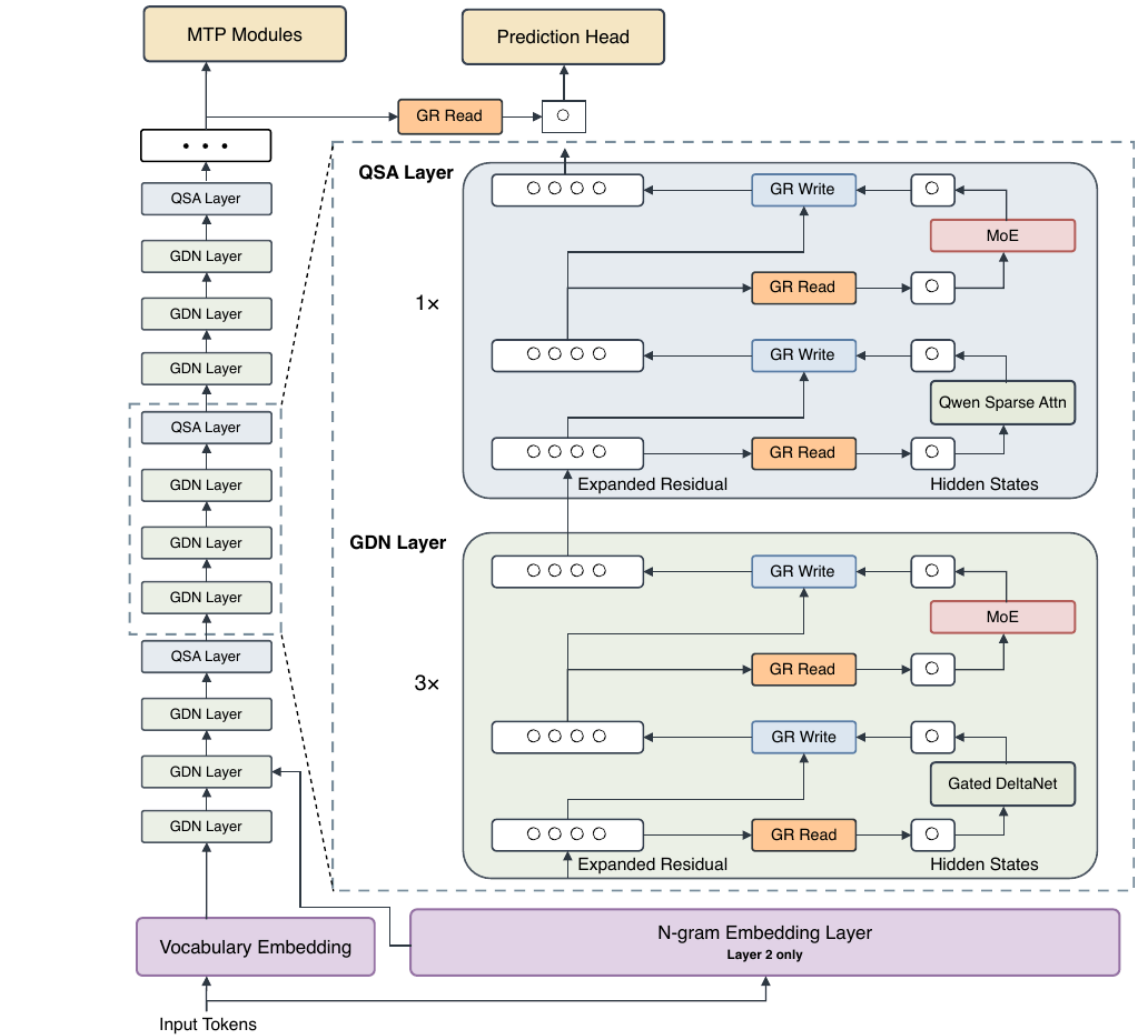
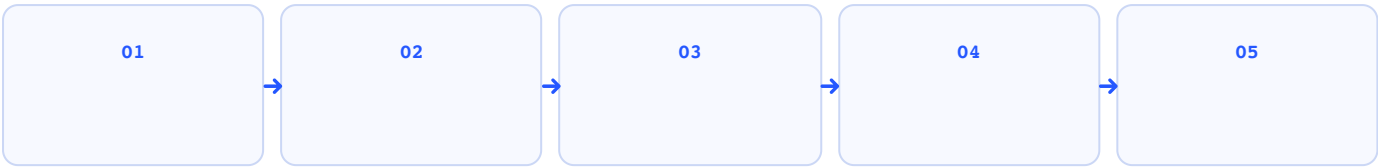


Figure 1: **Qwen3.8-Flash-Next architecture.** Token mixing alternates three GDN layers with one QSA layer per block of four. Every sublayer reads and writes through GR, which widens the residual stream and gates the read elementwise. An n-gram embedding layer at Layer 2 scales capacity off the accelerator via host-memory prefetching. The MTP module reuses QSA indices across speculative decoding steps.

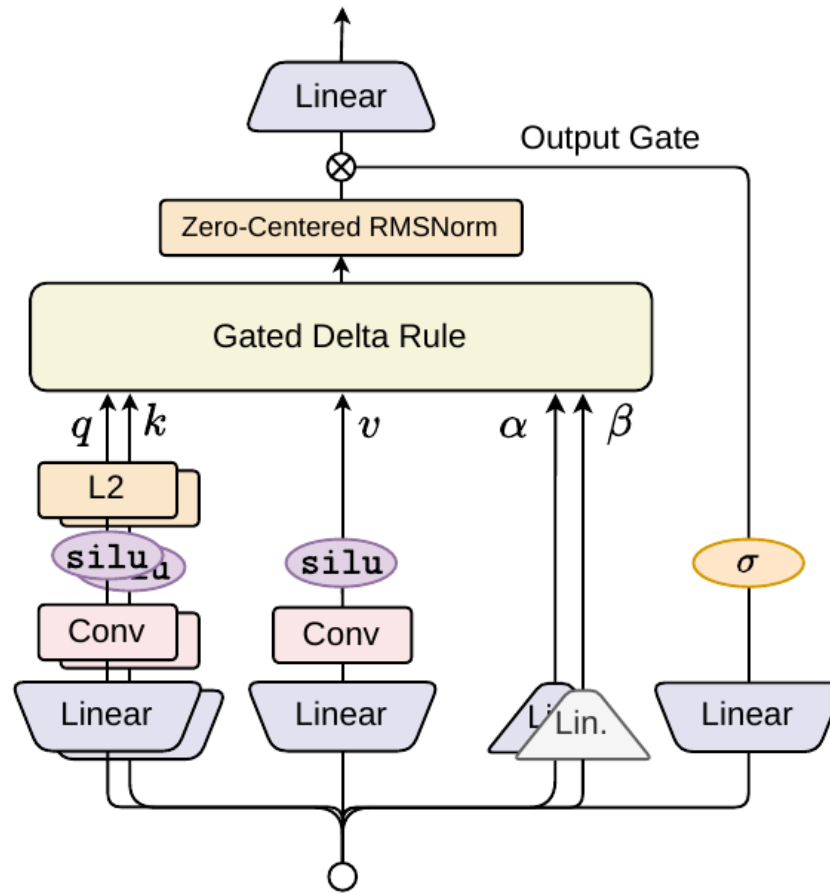


03 · GATED DELTANET

GDN 不是“缩小窗口的 attention”。它给每个 head 维护一个固定大小的 fast-weight 矩阵，当前 token 用 key 定位旧关联，用 value 与旧读出之差做增量写入，同时通过 decay gate 控制整张记忆的寿命。

tion

N Hybrid Architecture



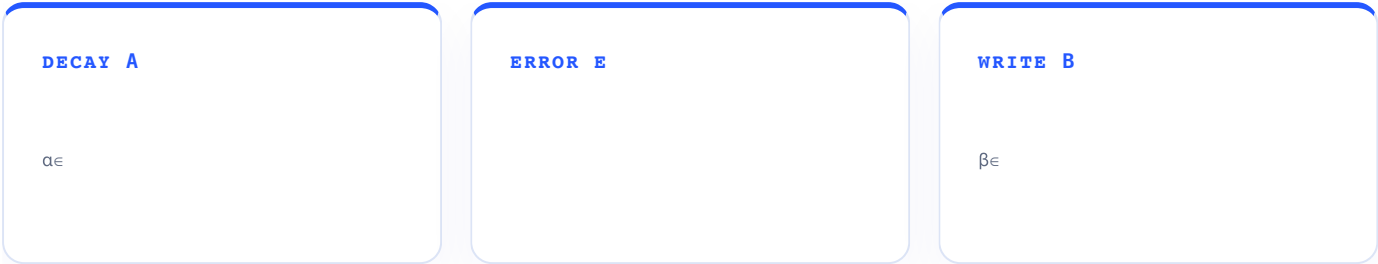
The Gated DeltaNet token mixer. The projected query, key, and value streams parallel convolutions; queries and keys are L2-normalized before the gated delta recurrent update. α_t and write gate β_t control the recurrent update, while a sigmoid output gate modulates the RMS-normalized output.

$$\tilde{s}_{t-1} = \alpha_t s_{t-1}$$

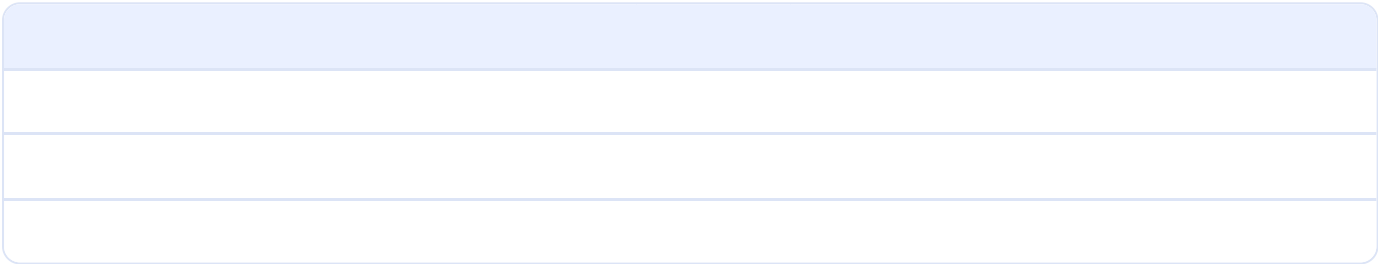
$$e_t = v_t - \tilde{s}_{t-1}^T k_t$$

$$s_t = \tilde{s}_{t-1} + \beta_t k_t e_t^T$$

$$y_t = s_t^T q_t$$



-
-
-
-



04 · QWEN SPARSE ATTENTION

QSA 明确沿着 DeepSeek Sparse Attention 的“轻量 indexer → Top-K → sparse core attention”路线继续设计。它针对 DSA 的剩余瓶颈做了关键改动：先把 key 序列按 4-token micro-block 压缩，再在 block 级打分，将索引复杂度从 $O(n^2)$ 降为 $O(n^2/r)$ 。

computational bottleneck of softmax attention in long-context scenarios. Recently, DSA (Liu et al., 2023a) has achieved considerable inference speedups using a lightweight indexer to generate token-level sparse masks. However, as sequence length increases, the overhead of its $O(n^2)$ indexer remains non-negligible.

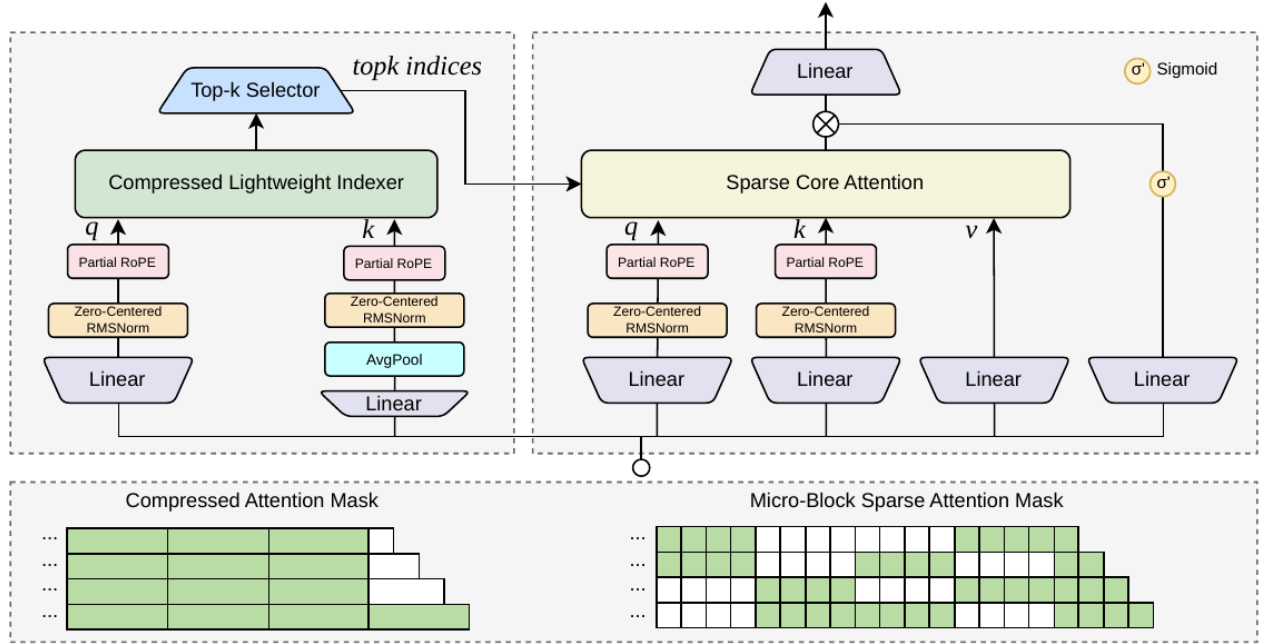
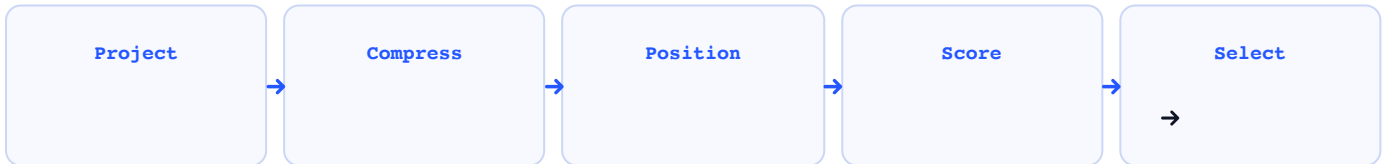


Figure 3: Overview of Qwen Sparse Attention (QSA). The QSA indexer (left) uses a compressed causal attention mask to score key blocks and select the top- k indices. These indices are expanded into a micro-block sparse attention mask for sparse core attention (right).



$$\bar{k}_b = \text{RMSNorm}(\text{AvgPool}(k_{p:p+r-1}))$$

$$I_{ib} = \sum_{h=1 \dots H} \text{ReLU}(\langle q_{ih}, \bar{k}_b \rangle), \quad \text{block}$$

$$K_B = \lfloor K/r \rfloor = 512 \quad K=2048 \quad r=4$$

```
# Stage 1: dense distillation; backbone teacher          attention
teacher_token = ll_normalize(sum_over_heads(softmax(full_attn_logits)))
teacher_block = ll_normalize(max_pool(teacher_token, block_size=4))
loss = KL(teacher_block || softmax(indexer_score))
train(indexer_only, steps=1000, lr=1e-3, tokens≈2B)

# Stage 2: sparse CPT; Top-K
blocks = topk(indexer_score, 512)
tokens = expand(blocks) + causal_tail
output = sparse_core_attention(q, k[tokens], v[tokens])
train(backbone_and_indexer, steps=8000, lr=2.5e-5, tokens≈200B)
```

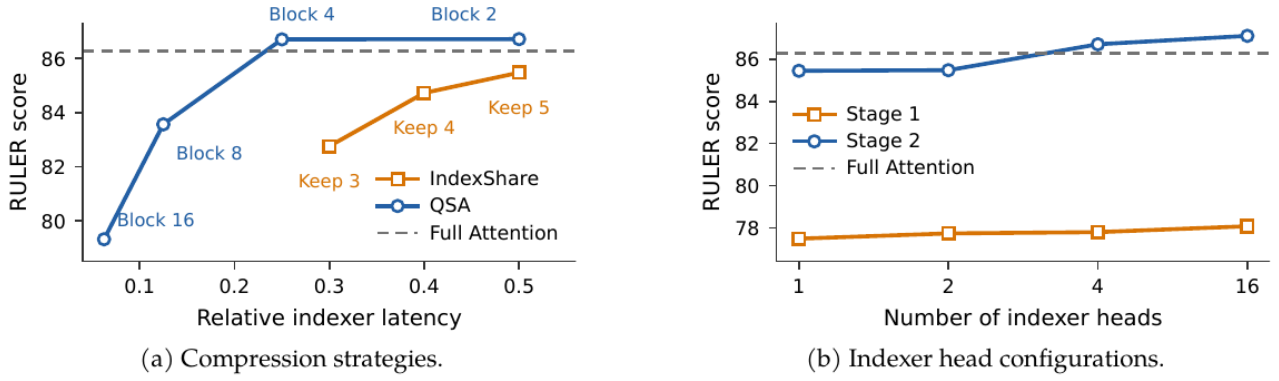


Figure 5: Architecture ablations of QSA on RULER. (a) QSA performance with different micro-block sizes; “Keep x ” indicates the number of IndexShare indexer layers retained for computation. (b) Performance with different numbers of indexer query heads after dense distillation and sparse training.

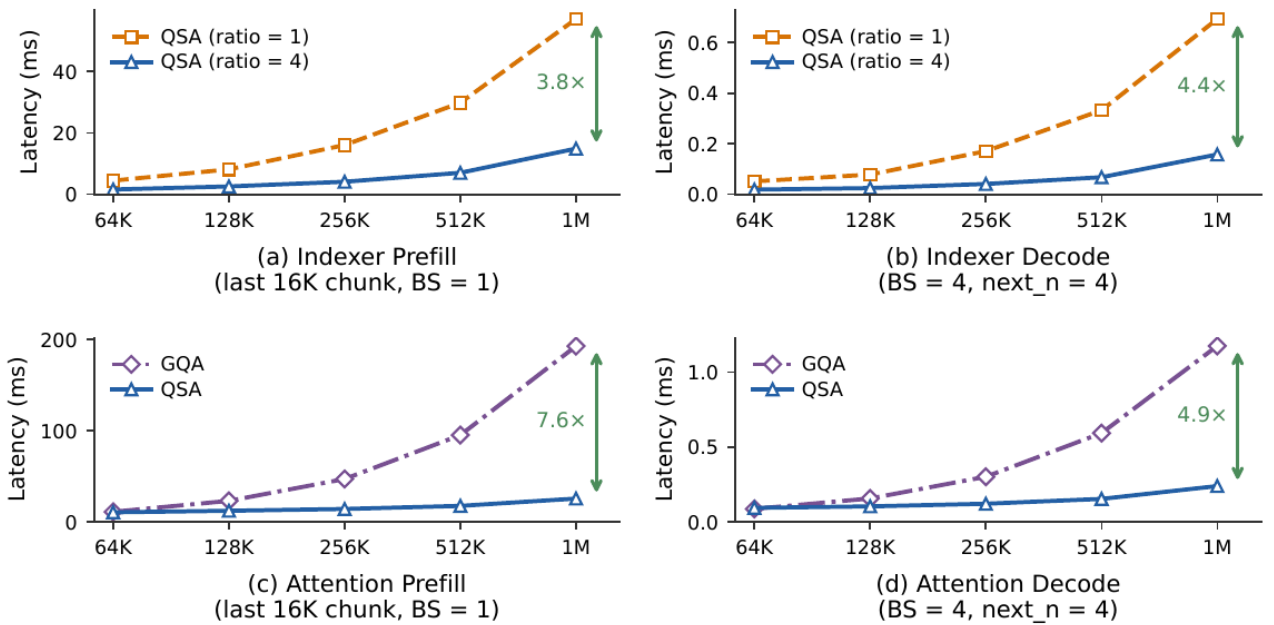
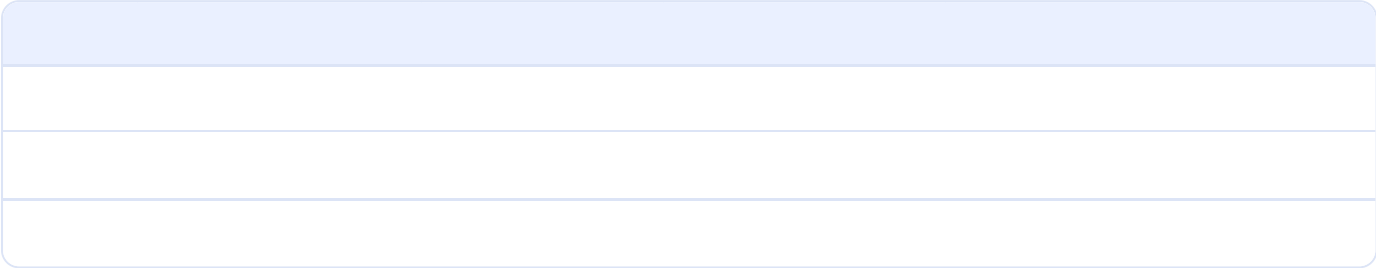


Figure 6: Kernel-level latency of QSA across context lengths during prefill and decode. Panels (a,b) compare indexer latency under different compression ratios, while panels (c,d) compare kernel-level attention latency between dense GQA and QSA, including both the indexer and sparse core attention. Chunked prefill uses a 16K-token chunk with batch size 1; decode uses batch size 4 and next_n = 4, corresponding to three MTP prediction steps. Arrows indicate speedups at a context length of 1M.

x x

x x

普通 pre-norm residual 让所有层读写同一个向量。早期特征必须与后续所有输出叠加在同一条流里。GR 把残差状态扩成四个平行分支，并让每个子层按 token、按通道决定读多少，再用每分支一个标量决定写多少。



```

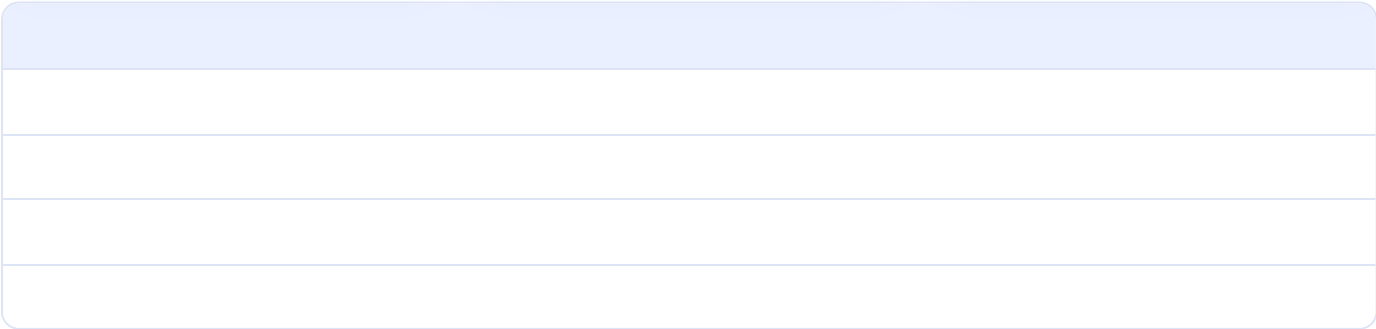
$$\hat{R}_i = \text{RMSNorm}(R_i; Y_i)$$

$$G = \text{unvec}(\sigma(W_u \text{ SiLU}((1/n_r)W_d \text{ vec}(\hat{R}))))$$

$$x = (1/n_r) \sum_i G_i \odot \hat{R}_i$$

$$s = 2\sigma((1/n_r)W_w \text{ vec}(\hat{R})), \quad R'_i = R_i + s_i Y$$

```



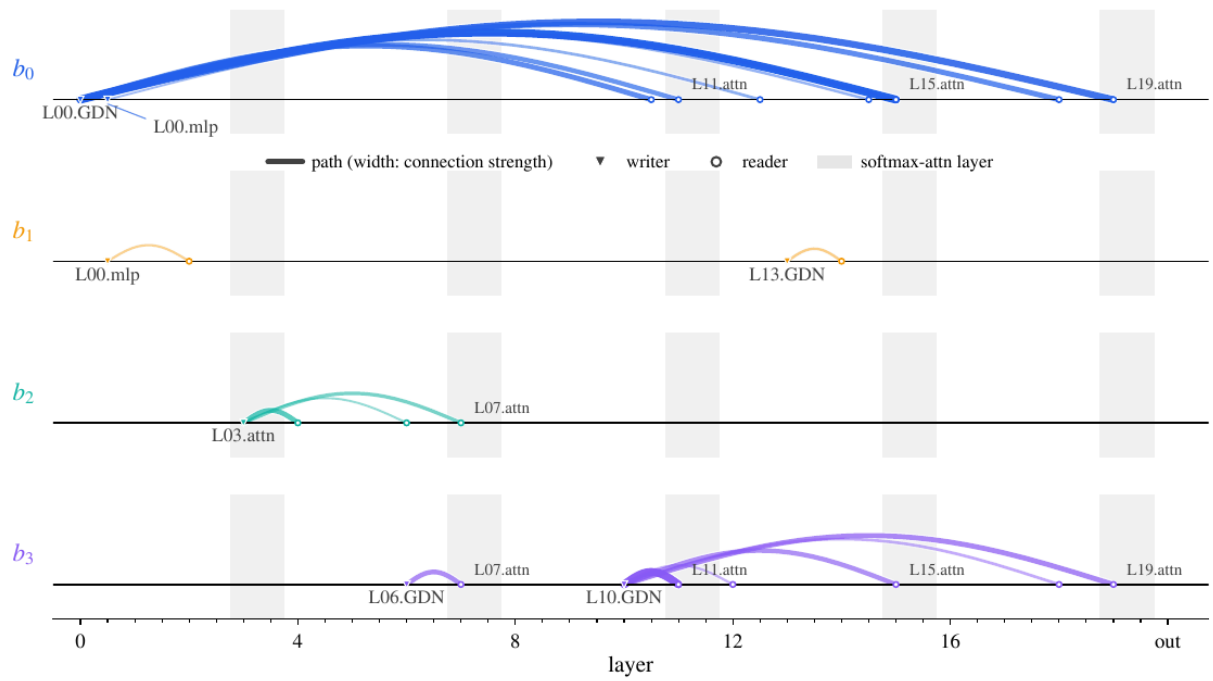


Figure 7: **Cross-layer paths added by GR.** Each row corresponds to one residual branch; a connection runs from the sublayer that wrote into that branch to a later sublayer that reads it back. Vertical position encodes the number of layers skipped; line width and opacity encode Δ_{uv} (Eq. (37)), the additional share of the reader’s input supplied by this writer compared to a single residual stream. Shaded regions denote softmax-attention layers, every fourth in this hybrid; the rest are GDN, and sublayers are named accordingly (L00.GDN, L03.attn, L00.mlp). Readers are named where they are softmax-attention sublayers, which is where the long-range paths land. One branch carries long-range paths (all connections on b_0

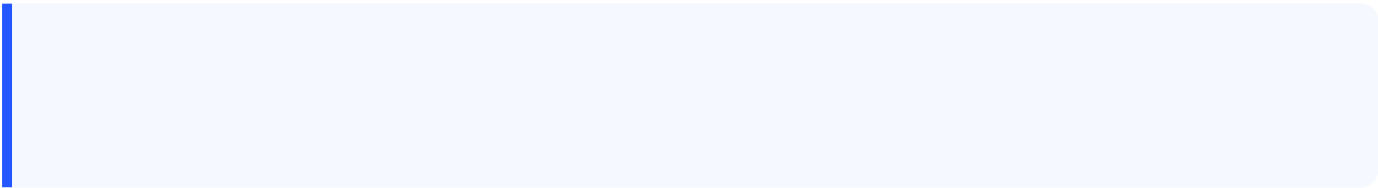
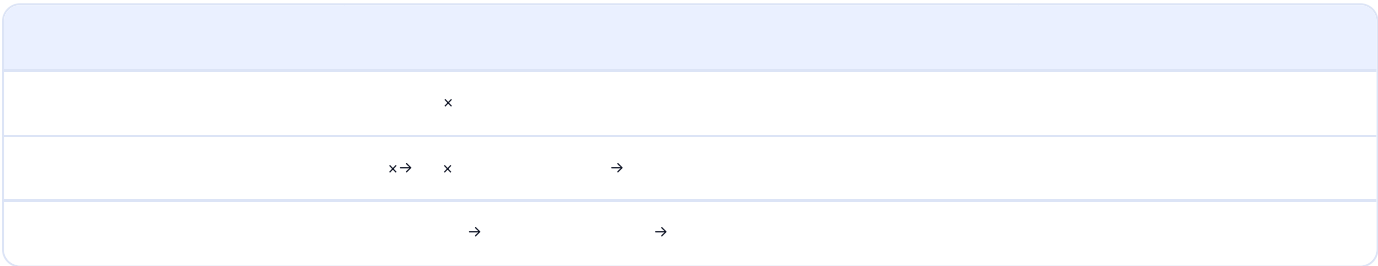
~

~

→

→

MoE 用 router 选择要计算的专家；n-gram memory 用最近的局部 token 片段直接生成地址，查出向量并加到 token representation。前者增加条件计算，后者增加条件记忆，两者是正交的稀疏化轴。

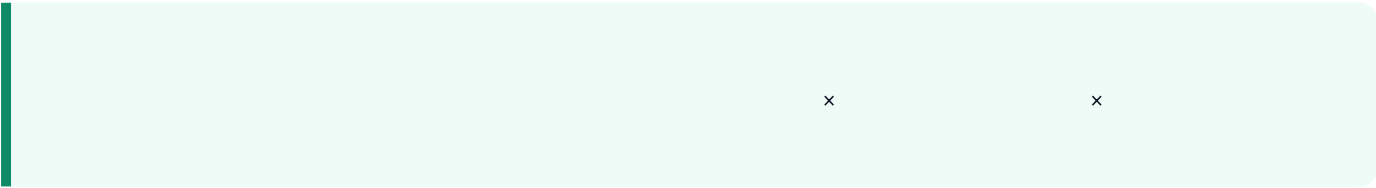


07 · OPTIMIZER & SYSTEM

Muon 对矩阵动量做 Newton – Schulz 正交化。真正困难的是：Megatron 中一个逻辑矩阵可能被 TP 切开，而不同矩阵的 NS FLOPs 随短边近似三次增长，按参数元素平均切 DP 会产生严重 straggler。Qwen 通过 Canzona 解耦“物理存放位置”和“谁负责算 optimizer”。

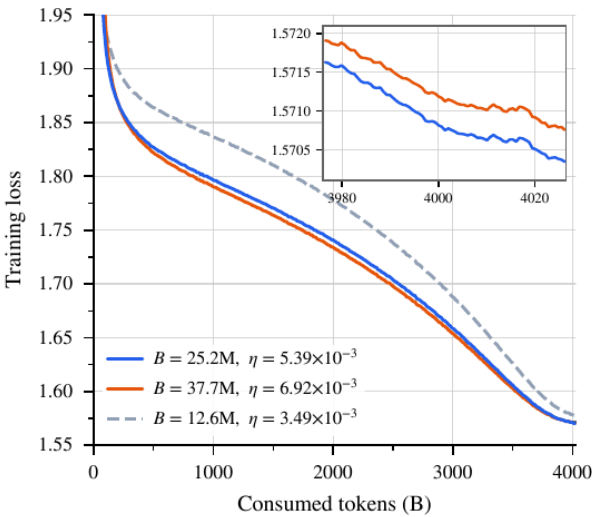
$\alpha \beta$

- μ
-
-

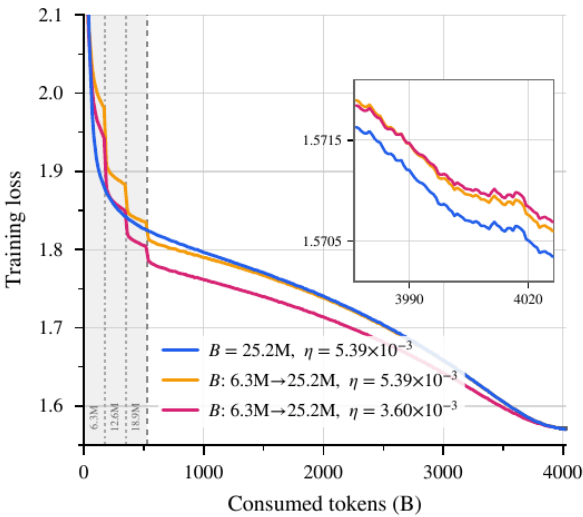


08 · HYPERPARAMETER SCALING

旧 Qwen3.5 的 scaling law 不能直接平移。团队重新拟合 η 与 batch size：新配方的近优区域整体向更大 batch、更高 learning rate 移动，而且大 batch 的损失曲线在最优点上方很平，在下方却陡峭。



(a) Constant batch size.



(b) Ramped batch size.

Figure 8: **Batch size on a 4T-token budget.** Training loss against consumed tokens for a 20-layer 10.8B-A0.89B MoE; each inset resolves the last 50B tokens. (a) Moving from the previous recipe ($B = 12.6\text{M}$) to the predicted optimum ($B = 25.2\text{M}$) is worth 7.2×10^{-3} , while a further step to $B = 37.7\text{M}$ leads to a minor degradation. The loss rises steeply below the prediction and remains nearly flat above it. (b) Reaching $B = 25.2\text{M}$ through a ramp instead performs no better than using this batch size from the start and takes 18.8% more optimizer steps.

learning rate (η) and batch size (B) (Qwen Team, 2024; Yang et al., 2025). The optimal learning rate and batch size depend on the model architecture and the optimizer, so a recipe that was optimal for the previous-generation model may become suboptimal once both change. Under our previous Owen3.5 hv-

	→
η	-
η	-

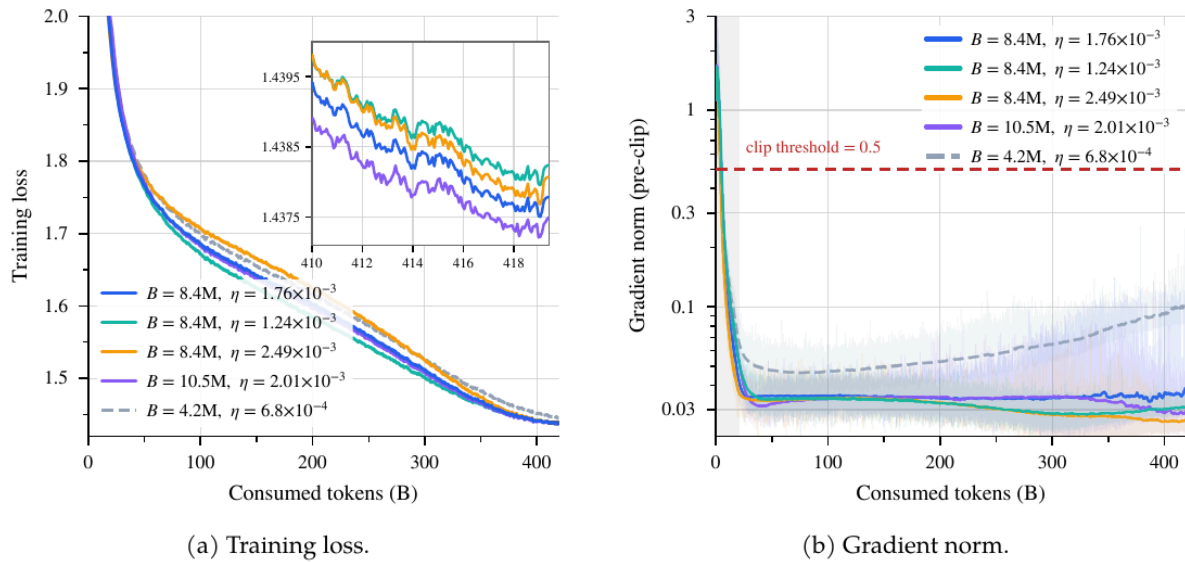


Figure 9: **Learning rate at a larger model scale.** Five runs of a 48-layer MoE on a 419B-token budget: the predicted optimum ($B = 8.4M, \eta = 1.76 \times 10^{-3}$), η divided and multiplied by $\sqrt{2}$, a 25% larger batch with its matched η , and the previous recipe (dashed). (a) The inset covers the last 10B tokens. The loss differences among the predicted optimum and the three nearby settings are near the noise level, so the predicted optimum sits at the bottom of a flat bowl. (b) Pre-clip gradient norm on a log scale, with individual steps faint behind a moving average, the warmup shaded and the clip threshold dashed. After warmup, the pre-clip gradient norms of all runs near the predicted optimum remain below 50% of the clipping threshold, including the run at $\sqrt{2}$ times the predicted learning rate.

Learning Rate at a Larger Model Scale. The learning-rate prediction is tested on a much larger 48 layers 156B-A7B MoE, over a 419B-token budget that is again the same for every run. We evaluate the predicted optimum together with learning rates scaled by $1/\sqrt{2}$ and $\sqrt{2}$, a 25% larger batch with its matched learning rate, and the previous hyperparameter recipe

 $\pm\sqrt{}$
 $\times^{-4}$

09 · TRAINING STABILITY

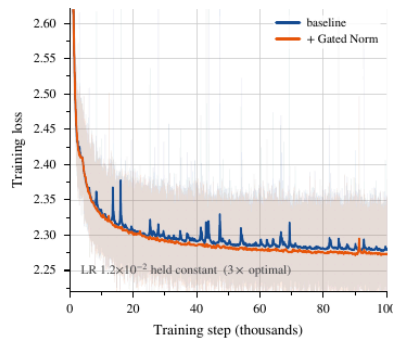
大模型生产训练的偶发 spike 很难用小实验观察。报告采用 stress test：在 28-layer MoE 上保持 $2\times$ 、 $4\times$ 最优学习率不衰减，让本应在超长生产训练中才出现的不稳定性在 10 万步内暴露。

$2 \times \text{LR}$ $4 \times \text{LR}$

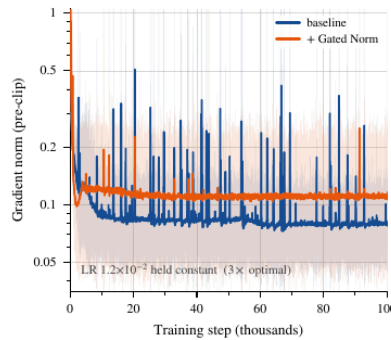
GATE ISOLATION

→
→

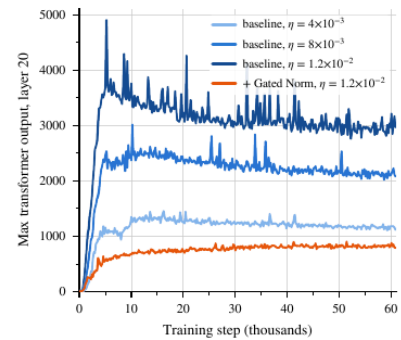
(Fig. 10). The AdamW run spikes on 183 per 10k steps and crosses the clipping threshold on 213 of 19,932 steps, engaging the clipper continuously. In contrast, both Muon runs never cross the clipping threshold, and the configuration with the gated residual records zero loss spikes. Under equal stress, the new architecture and optimizer combination is substantially more stable.



(a) Training loss.

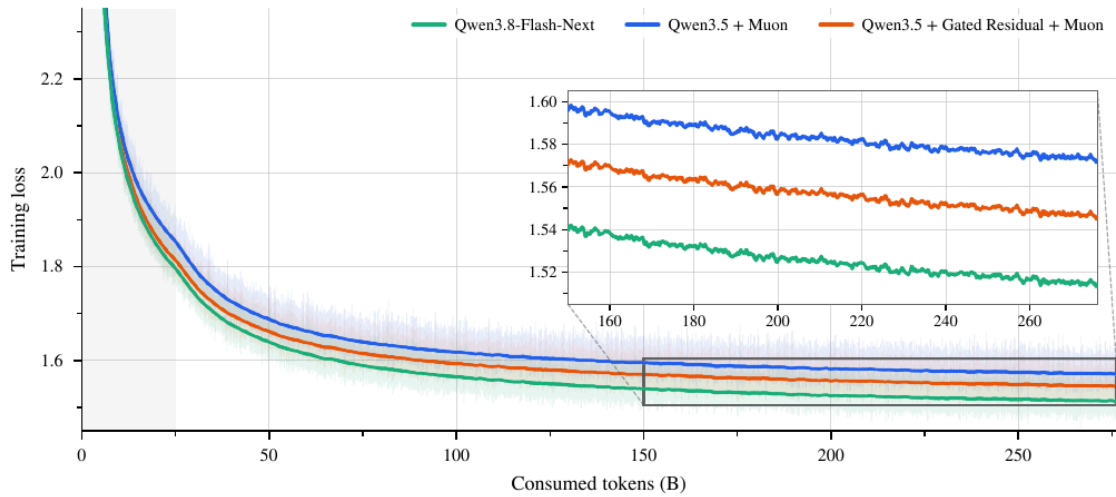


(b) Pre-clip gradient norm.

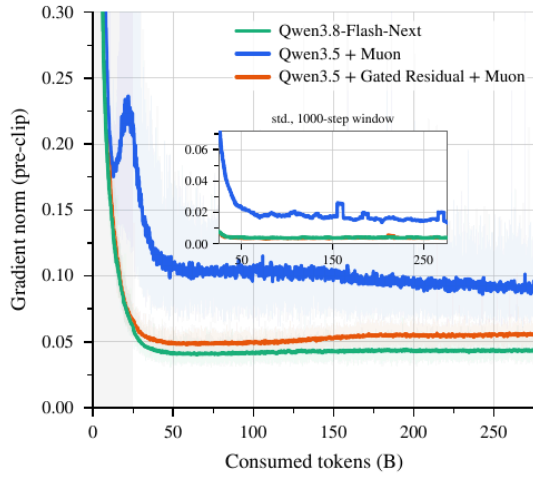


(c) Outliers against learning rate.

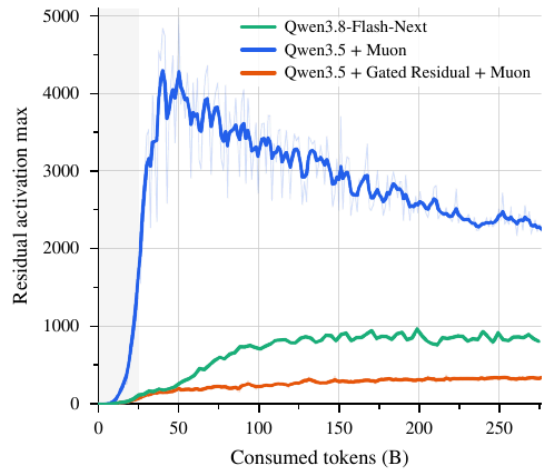
Figure 12: **Isolating the effect of the gate.** GatedNorm off and on, with the AdamW optimizer, structure and data order held fixed. (a) and (b) are the pair at $3 \times$ the optimal learning rate; bold lines are a moving average over the faint per-step trace. (c) adds the ungated baseline at $1 \times$ and $2 \times$ the optimal rate.



(a) Training loss.



(b) Pre-clip gradient norm and sliding window std.



(c) Residual maximum over training.

Figure 13: **The early phase of Qwen3.8-Flash-Next, at the shipped learning rate.** The first 276B tokens of three runs that share data order, learning-rate schedule, and optimizer: Qwen3.5 with Muon, the same plus the gated residual, and Qwen3.8-Flash-Next. The shaded band corresponds to the learning-rate warmup. The inset in (a) resolves the last 126B tokens of the window; the inset in (b) is the standard deviation of the gradient norm inside a rolling 1000-step window.

•
•
•

×

10 · DISCLOSURE AUDIT

这份报告的重点是架构实验，不是完整训练或部署白皮书。它披露了大量 ablation token budget、QSA CPT token 和相对成本，但没有给出主模型总训练 token 的绝对值、训练卡数、硬件拓扑、墙钟时长或 MFU。

≈ ≈

×	×
---	---

11 · EVALUATION

Table 11 比较 Qwen3.8-Flash-Next-Base、Qwen3.8-27B-Base 与 Qwen3.7-Plus-Base。Flash-Next 在 14 项中全部超过 27B dense，并在 8/14 超过 397B-A17B 前代；其余差距最大 2.6 分。

- **Multilingual tasks:** MGSM (Sui et al., 2022) (8-shot, Co1), MMMLU (OpenAI, 2024) (3-shot), INCLUDE (Romanou et al., 2024) (5-shot).

Table 11: Comparison among the base models of Qwen3.8-Flash-Next, Qwen3.8-27B and Qwen3.7-Plus. The highest and second-best scores are shown in **bold** and underlined, respectively.

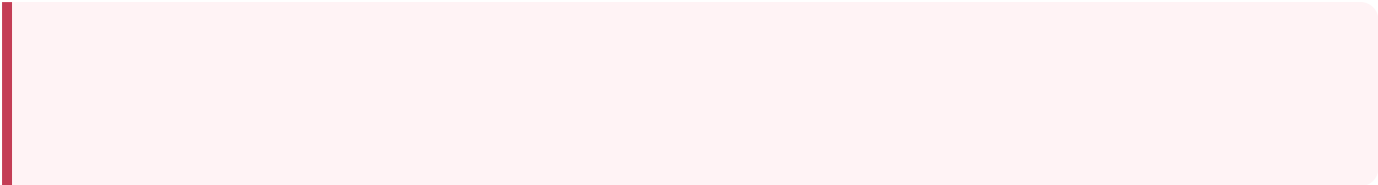
	Qwen3.8-Flash-Next-Base	Qwen3.8-27B-Base	Qwen3.7-Plus-Base
# Params	125B	27B	397B
# Activated Params	6B	27B	17B
# N-gram Embedding Params	51B	–	–
General Tasks			
MMLU	<u>90.36</u>	87.51	90.43
MMLU-Redux	<u>90.68</u>	87.26	91.47
MMLU-Pro	73.23	68.60	<u>70.90</u>
SuperGPQA	51.36	44.86	<u>48.42</u>
BBH	90.87	89.56	<u>89.41</u>
Math & STEM Tasks			
GPQA	<u>51.42</u>	45.01	51.52
GSM8K	93.29	<u>93.18</u>	92.95
MATH	<u>72.78</u>	60.54	74.38
Coding Tasks			
EvalPlus	78.76	76.05	<u>78.06</u>
MultiPL-E	<u>79.09</u>	74.50	81.68
SWEBench-Pretrain	50.99	41.66	<u>49.24</u>
Multilingual Tasks			
MGSM	89.33	<u>86.37</u>	85.42
MMMLU	84.86	<u>79.74</u>	<u>84.53</u>
INCLUDE	<u>78.40</u>	74.37	78.90

VS 27B DENSE

VS 397B-A17B

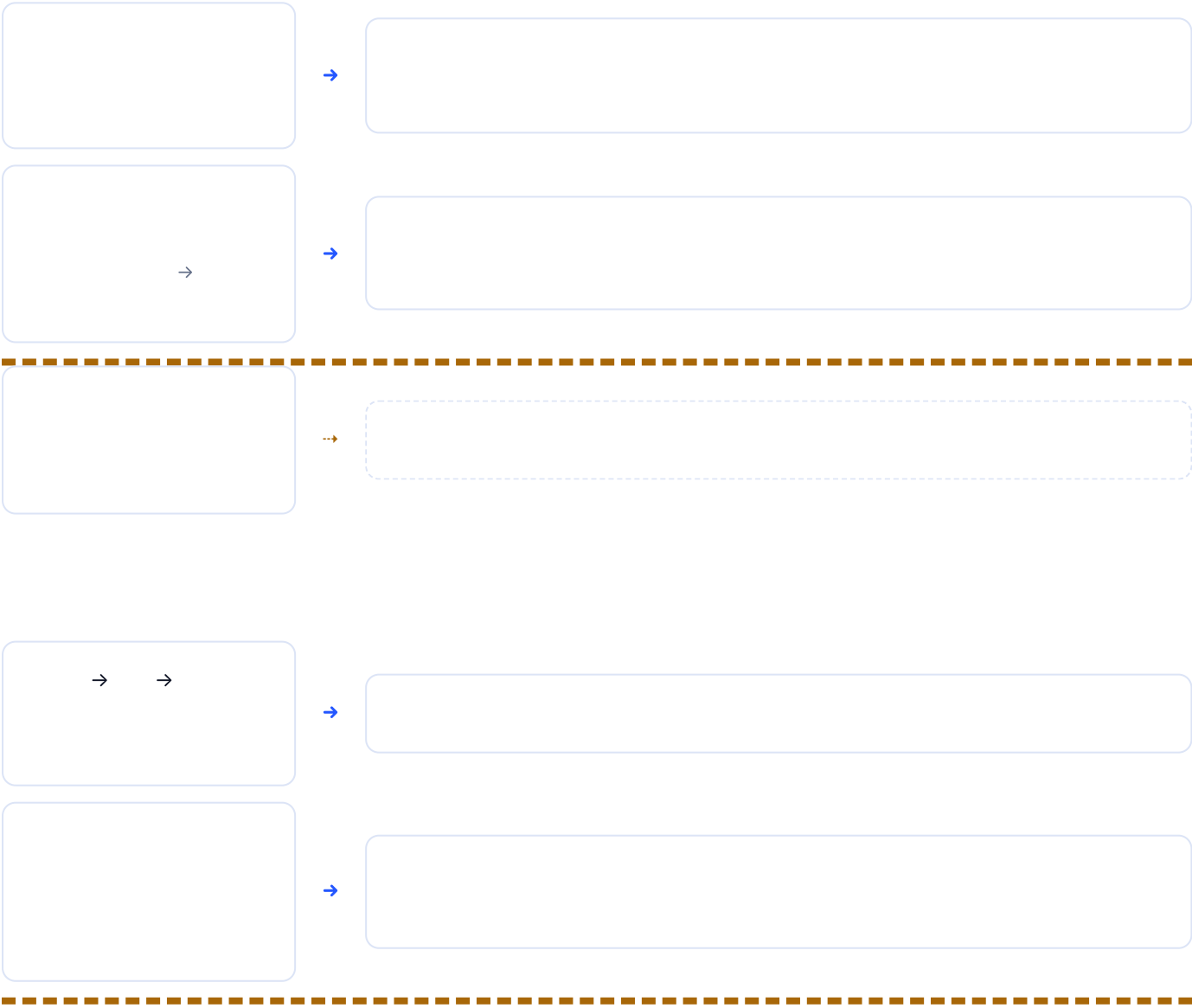
COST CLAIM

≈



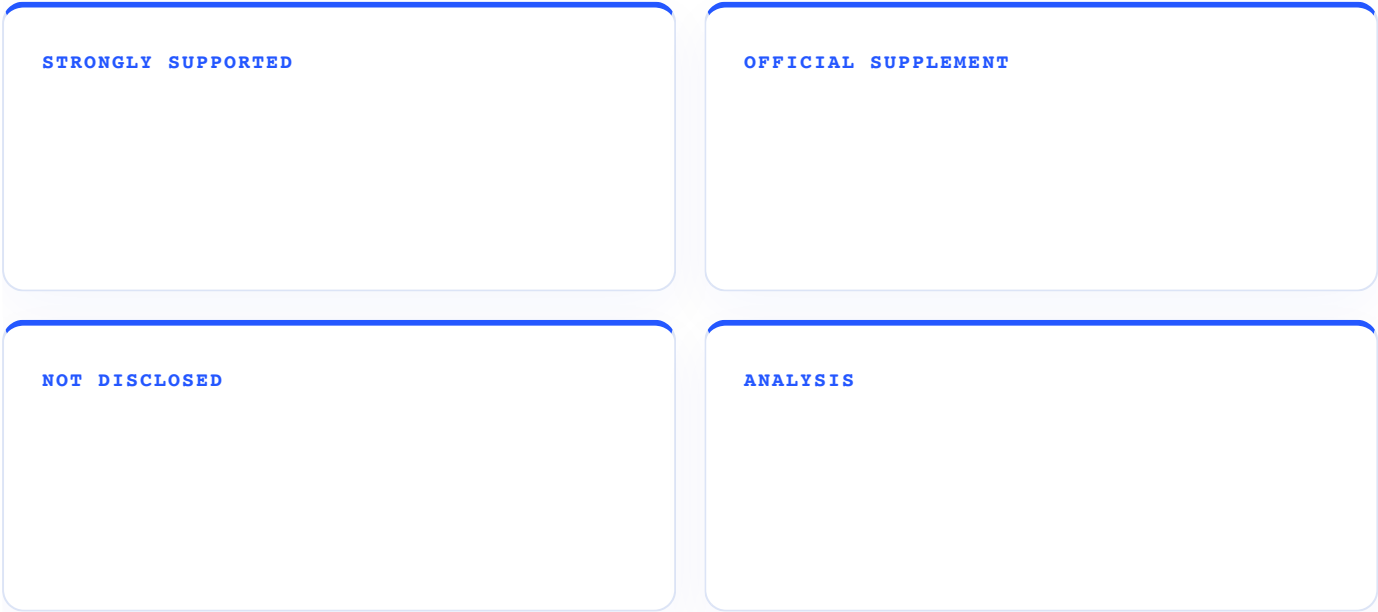
12 · TECHNICAL PROVENANCE

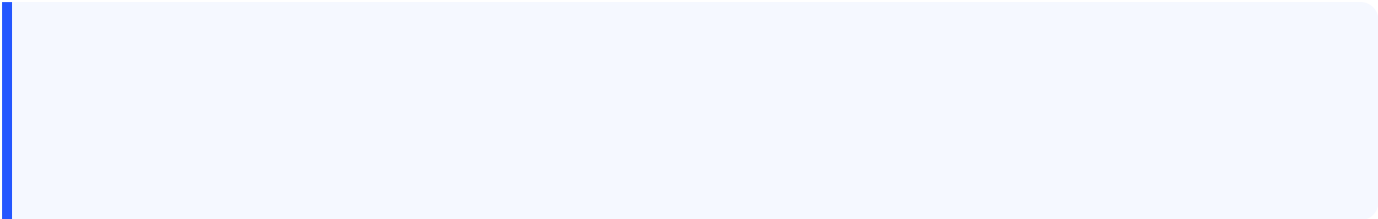
下图按证据强度描述技术关系。实线节点表示报告明确采用或组合；虚线节点表示同问题域的结构比较。相近不自动等于来源。





13 · EVIDENCE BOUNDARIES





14 · PRIMARY SOURCES

01

02

03

04

05

06

07

08

09

10

11

12

13

